METRIC

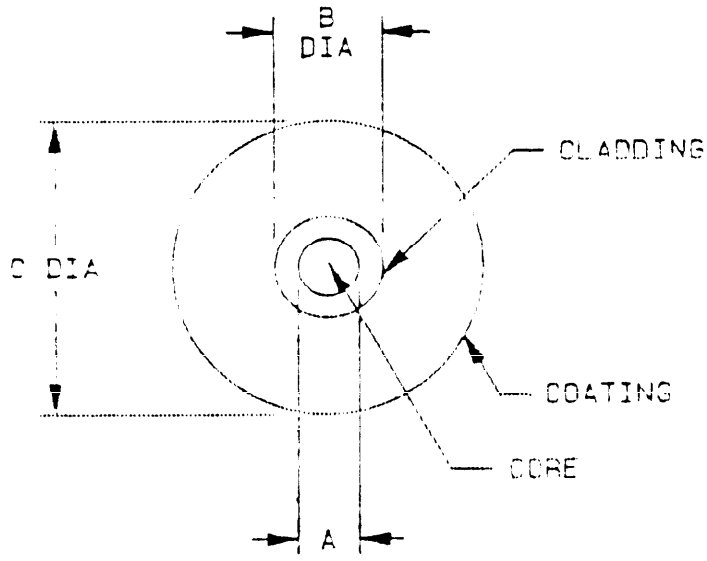
MIL-F-49291/09
21 May 1990

MILITARY SPECIFICATION SHEET

FIBER, OPTIC, 400/430 µm, RADIATION HARDENED, AVIONIC RATED (METRIC)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-F-49291.



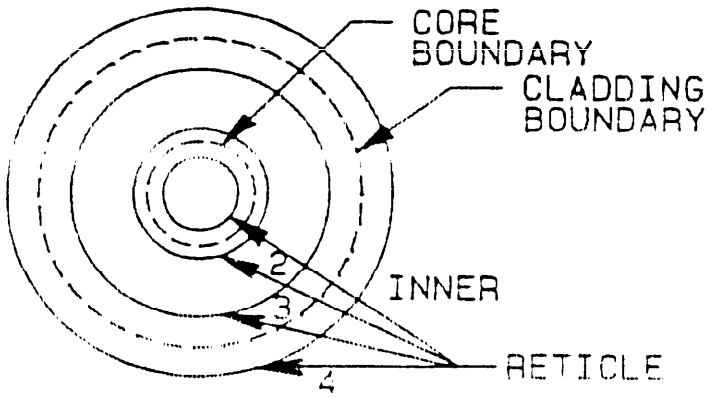
Part or Identifying Number	Dimensions (µm)		
	A	B	C
M49291/09-01	400 ±5	430 + 0 -10	730 ±50

NOTE: Dimensions are in micrometers.

FIGURE 1. Dimensions and configuration of primary coated optical fiber construction.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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Circle	Diameter of circle
Inner	395 µm
Second	405 µm
Third	420 µm
Fourth	430 µm

NOTE: Dimensions are in micrometers.

FIGURE 2. Tolerance fields.

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OR REMOVE THE TEST NUMBER FROM THE NUMBER SET.

REQUIREMENTS:

Classification:
Type: I
Class: 3
Composition: B
Style: 3

Dimensions and configuration:
Diameter: see figures 1 and 2.
Ovality:
Core: 5 percent
Cladding: 4 percent
Coating: 10 percent
Offset:
Core-to-cladding: 7 µm
Fiber-to-coating: 35 µm
Maximum percent of coating diameter change at the splice points: Not applicable.

Splices: Not allowed.

Tensile strength (proof test): 690 MPa

Fiber mass/unit length (kg/km): 900 grams maximum/ km

Maximum attenuation rate: 6 dB/km, 850 nm ±10 nm full width half maximum (FWHM).
After nuclear event: Maximum allowable degradation 24 hours after exposure shall not exceed 10 dB/km at 25°C.

Numerical aperture: 0.37 ±0.02 at 850 nm.

Bandwidth: 13 Mhz-km (850 nm ± 10 nm).

Fluid immersion: Not applicable.

Temperature range:
Operating: -65°C to +125°C.
Storage: -65°C to +125°C.

Part or Identifying Number (PIN): M49291/09-01.

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Qualification inspection shall consist of performing the inspections and optical test specified in table I herein, in the sequence shown.

Group A inspection shall consist of the inspections and optical test specified in table II herein, in the sequence shown.

Group B inspection shall consist of the inspections and optical test specified in table III herein, in the sequence shown.

Group C inspection shall consist of the inspections and optical test specified in table IV herein, in the sequence shown.

TABLE I. Qualification inspection.

Inspection	Test sample length	Sample quantity
Group I		
Visual and mechanical	1 km	10 units
Attenuation rate	1 km	10 units
Numerical aperture	2 m	10 units
Core diameter	2 m	10 units
Bandwidth by time domain	1 km	10 units
Group II		
Proof test	1 km	5 units
Mechanical strippability	2 m	5 units
Group III		
Temperature cycling	1 km	5 units
Temperature shock	1 km	5 units
Temperature-humidity cycling	1 km	3 units
Dimensional stability	5 m	3 units
Nuclear radiation resistance	50 m minimum	3 units 1/
Fungus	5 m	2 units
Life aging	1 km	2 units

1/ One unit each from 3 fibers.

TABLE II. Group A inspection.

Inspection	Test sample length
Visual and mechanical	1 km
Attenuation rate	1 km
Numerical aperture	2 m
Core diameter	2 m
Bandwidth by time domain	1 km

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to reference jumper using ... Step 3 - Connect test jumper

TABLE III. Group B inspection.

Inspection	Test sample length
Proof test	1 km
Mechanical strippability	2 m

TABLE IV. Group C .

Inspection	Test sample length	Sample quantity
Subgroup I (3-month intervals)		
Temperature cycling	1 km	4 units
Subgroup II (6-month intervals) interval)		
Temperature shock	1 km	2 units
Temperature-humidity cycling	1 km	2 units
Subgroup III (12-month intervals)		
Dimensional stability	5 m	2 units
Nuclear radiation resistance	50 m minimum	3 units 1/
Fungus	5 m	2 units
Life aging	1 km	2 units

1/ One unit each from 3 fibers.

CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - EC
Air Force - 85

Review activities:
Army - MI, SC
Navy - AS
Air Force - 11, 71
DLA - ES

User activity:
Air Force - 17

Preparing activity:
Air Force - 85

Agent:
DLA - ES

(Project 6010-0031)

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